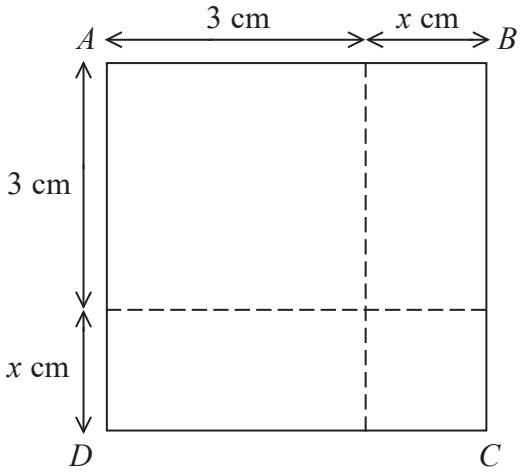
**4**




The area of square $ABCD$ is 10 cm$^2$.

Show that $x^2 + 6x = 1$

Each side is $x + 3$, so $(x + 3)(x + 3) = 10$.

| × | $x$ | $+3$ |
|---|---|---|
| $x$ | $x^2$ | $+3x$ |
| $+3$ | $+3x$ | $+9$ |

**1 mark**
Express the square area

$x^2 + 3x + 3x + 9 = 10$

**1 mark**
Expand and use area 10

$x^2 + 6x + 9 = 10$

$-9$ $\quad$ $-9$

$x^2 + 6x = 1$

**1 mark**
Subtract 9 from both sides

**(Total for Question 4 is 3 marks)**